

A formal proposal for a MSc project that will be submitted in partial fulfillment of a University of Greenwich Master's Degree

Insertion of Data Mining Techniques with Feature Selection to Predict Share Price on Stock Exchange Data

Name: Vennela Yaganti

Student ID: 001413388

Course of Study: MSc Data Science

Date Proposal Submitted : 09-03-2025

Submission Date: 15-03-2025

Supervisor: Simon Scola

Topic Area: Insertion of Data Mining Techniques with Feature Selection to Predict Share Price on Stock Exchange Data

Keywords: Stock Price Prediction, Data Mining Techniques, Feature Selection, Machine Learning Models, Historical Stock Data, Time Series Analysis, Predictive Analytics, Web-Based Deployment

MSc Modules studied that contribute towards this project:

Machine Learning, Big Data, Statistical Methods for Time Series Analysis, Programming Fundamentals for Data Science, Applied Machine Learning, Data Visualization, Technologies for Anti-Money Laundering.

1. Overview

The research will make use of data mining techniques and feature selection methods to develop a predictive model of daily historical stock prices of companies in the United Kingdom. However, predicting the stock market is still a complicated task because of the volatile market and the effect of many factors. Using data mining algorithms as a base, the model will incorporate feature selection to determine the most relevant feature that affects stock prices. The study will use open-source UK stock market data of various companies from Kaggle with the help of a secondary data collection strategy. Different data mining techniques will be used in analysing historical data to choose the most important predictor. Finally, Python programming language will be used to develop machine-learning models for prediction to enhance the forecasting performance. The model will then be implemented in a web interface for user accessibility. MSc Data Science is adopted as a project theme with principal components from data preprocessing, feature selection, predictive modelling and deployment. Data cleaning, exploratory data analysis, and algorithm selection techniques will be hinted at along with their model evaluation will be performed in accordance with QCF protocol. By combining a data-driven approach to UK stock market price prediction, this study can improve the decision-making process of investors and analysts in financial analytics.

2. Objectives

- To gather and prepare stock exchange data from open-source resources to ensure data quality and consistency for stock price forecasting using the Python programming language.
- To apply data mining techniques to analyse historical stock price trends and identify relevant patterns.
- To implement feature selection methods to determine the most significant factors influencing stock price movements.
- To build a predictive modelling approach using machine learning algorithms to forecast stock prices with improved accuracy.

- To evaluate the model's performance using statistical metrics to validate its predictive capability with data-driven insights.
- To deploy the model in a web interface for real-time stock price prediction and user accessibility.

Research Questions

Research Question 1:

Can feature selection techniques be used to establish the factors that have the greatest impacts on stock price fluctuations in the UK stock market?

Research Question 2:

Can time series analysis enhance the identification of trends and their effectiveness in the predictive models of the stock price forecasts?

Research Question 3:

How does data mining technique impact the performance of the machine learning models in prediction of stock price if data pre-processing is done by handling missing values and normalizing it?

Research Question 4:

To what extent can a web-based machine learning model help investors enhance the stock market accessibility and usefulness in making better decisions?

Research Question 5:

How do various data mining algorithms affect the effectiveness of stock price prediction models when forecasting the UK stock market data? How the objectives will be achieved.

2.1 Data Collection and Preparation

Activities:

- The research will collect the data of the stock price from the historical stock price data available on the open source platforms like Kaggle.

- As the data needs to be pre-processed, we will deal with missing values and make identical data consistent.
- First, the data will be processed for cleaning by Python libraries such as Pandas and NumPy.

Deliverables:

- Insertion of Python modules in the Python notebook.
- A pre-processed and cleaned dataset will be ready for analysis.

2.2 Data Mining Techniques

Activities:

- The study will use the Python libraries Seaborn and Matplotlib to explore and represent the data.
- A time series analysis will be done to find out any pattern and correlation in historical stock trend price.
- With various graphical outputs, the stock price trends and the relationships between variables will be represented.

Deliverables:

- Visualizations and statistical summaries of stock price trends.
- Develop an EDA report providing key insights, trends and patterns in the dataset.

2.3 Feature selection

Activities:

- The target variable will be selected to predict stock market movement based on independent variables.
- By training with sets of data, the most pertinent set of features impacting on stock prices will be selected.
- In this section, we will use python libraries such as Scikit learn and Statmodels to find the influential elements affecting the stock prices.

Deliverables:

- A refined dataset with optimized feature selection.
- Train and test sets of data will be ready to fit the models.

2.4 Model building

Activities:

- Using Scikit-learn, the research will develop regression models such as Linear Regression, SVM Regression and Random Forest Regression.
- Models will be fitted on the train sets and forecasting will be applied on the testing sets.

Deliverables:

- The models will be tested and predicted on the testing sets of data.
- Models are ready for identifying predictive accuracy.

2.5 Model evaluation**Activities:**

- R-squared (R^2), Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) scores will be developed for trained regression models.
- Cross-validation methods will be involved to confirm model robustness.

Deliverables:

- The performance of the model will be compared with the model performance metrics.
- Insights into model accuracy and limitations will be provided.

2.6 Web interface deployment

- For the web interface of the modelling outcomes, the best model performance results will be executed by deploying relevant Python frameworks.
- Users will be able to input the data and get real-time stock price predictions using the web interface.

Deliverables:

- A web interface that will predict stock prices based on the input will be developed.
- It will give the documentation of deployment steps and user guides.

3. Legal, Social and Ethical Issues

The analysis will be performed which will not make any confidential and secure demographic details, since its aim is financial analytics drawn from publicly available data. The data will be sourced from open sources thus no copyrighted material will be used and properly cited. The security of non-confidential data is ensured by the product being the one which adheres to accepted standards. It will not pose any health and safety risks to the project. The usability is considered with accessibility features to make it accessible to a wider range of users. By using publicly available data without personal information, the project will be following the Data Protection Act. After all, the web interface will be made accessible by ensuring that the Disability Discrimination Act is followed. Nonetheless, the project might influence financial decision-making, hence improving investment strategies. The study will guarantee legal compliance with data protection laws and ethics standards, while responsibility will be held in handling financial data.

4. Resources

Dataset link:

<https://www.kaggle.com/datasets/eren2222/uk-daily-historical-stock-market-1988-2024>

Software: Pycharm

Documentation: MS Word, MS Excel

5. Critical success factors

Critical Success Factors

The project will be successful or unsuccessful based on the fidelity of the stock market data and the features that get chosen, as well as the accuracy of the predictive model. Quality issues will be challenging for historical data. Data inconsistencies, algorithm selection as well as model performance limitations will be risks. Through all these data preprocessing, the use of multiple machine learning algorithms, and the use of evaluation metrics to determine whether or not the machine learning algorithm was accurate, these risks will be mitigated. The web

interface will be deployed and will necessitate robust coding and testing to ensure access to the user and functionality. Improvement in predictive accuracy and usability will be done in a continuous manner and through continuous performance assessment.

Risk Management Plan

Risks	Likelihood	Impact	Mitigation Strategy
Data quality issues (missing values, inconsistencies)	High	High	Data preprocessing using Pandas & NumPy to clean and standardize data.
Selection of inappropriate features	Medium	High	Use Recursive Feature Elimination (RFE) and statistical tests to optimize feature selection.
Poor model performance	High	High	Experiment with multiple ML algorithms and fine-tune hyperparameters.
Deployment challenges in web interface	Medium	Medium	Conduct thorough testing and use robust frameworks like Flask.
Ethical and legal concerns	Low	High	Adhere to data protection laws and use publicly available data.

6. Schedule

Name	Duration	Start	Finish
Topic Selection	17 days?	24/2/25 8:00 AM	18/3/25 5:00 PM
Identify Research Scope	2 days?	24/2/25 8:00 AM	25/2/25 5:00 PM
Literature Review	10 days?	26/2/25 8:00 AM	11/3/25 5:00 PM
Data Collection and Preparation	5 days?	12/3/25 8:00 AM	18/3/25 5:00 PM
Data Mining Techniques	17 days?	19/3/25 8:00 AM	10/4/25 5:00 PM
Perform exploratory data analysis (EDA)	6 days?	19/3/25 8:00 AM	26/3/25 5:00 PM
Conduct time series analysis	5 days?	27/3/25 8:00 AM	2/4/25 5:00 PM
Generate visualizations and statistical summaries	6 days?	3/4/25 8:00 AM	10/4/25 5:00 PM
Feature Selection	12 days?	11/4/25 8:00 AM	28/4/25 5:00 PM
Determine the most significant features	5 days?	11/4/25 8:00 AM	17/4/25 5:00 PM
Selecting Dependent and Independent variables	3 days?	18/4/25 8:00 AM	22/4/25 5:00 PM
Split dataset into training and testing sets	4 days?	23/4/25 8:00 AM	28/4/25 5:00 PM
Predictive Modelling	14 days?	29/4/25 8:00 AM	16/5/25 5:00 PM
Implement regression models	7 days?	29/4/25 8:00 AM	7/5/25 5:00 PM
Fit models on training data	4 days?	8/5/25 8:00 AM	13/5/25 5:00 PM
Evaluate initial predictions	3 days?	14/5/25 8:00 AM	16/5/25 5:00 PM
Model Performance Evaluation	14 days?	19/5/25 8:00 AM	5/6/25 5:00 PM
Calculate R-squared, MSE, MAE, and RMSE metrics	8 days?	19/5/25 8:00 AM	28/5/25 5:00 PM
Compare model results	6 days?	29/5/25 8:00 AM	5/6/25 5:00 PM
Web Interface Deployment	18 days?	6/6/25 8:00 AM	1/7/25 5:00 PM
Implement web interface to get predictive outcomes	12 days?	6/6/25 8:00 AM	23/6/25 5:00 PM
Integrate best-performing model	6 days?	24/6/25 8:00 AM	1/7/25 5:00 PM
Documentation & Final Report	20 days?	2/7/25 8:00 AM	29/7/25 5:00 PM
Make dissertation with relevant chapters	15 days?	2/7/25 8:00 AM	22/7/25 5:00 PM
Analyse results in results chapter of dissertation	3 days?	23/7/25 8:00 AM	25/7/25 5:00 PM
Compile and submit the final research report.	2 days?	28/7/25 8:00 AM	29/7/25 5:00 PM

Figure 1: Timeline

7. Your references

Kaur and Dharni (2022) will deliver an extensive review of data mining techniques used in forecasting the stock market, highlighting the importance of some of the challenges and the performance evaluation for different models which will aid in the choice and evaluation of predictive models for this project.

As for the purposeful exploration of data mining techniques and feature selection methods for stock price prediction, Peng et al. (2021) will offer a full analysis that will provide valuable insights into how model evaluation metrics and approaches should be considered to improve a model's prediction accuracy.

In this regard, Rouf et al. (2021) will delve into comprehensive research of machine (learning) approaches to stock market prediction for the successful development of predictive models that are as accurate as possible, paying attention to the data pre-processing, feature selection as well as the model appraisal.

Singh and Khushi (2021) will help to develop a technique for stock price prediction which has been created by combining a few technical and fundamental indicators.

In their feature ranking analyst ratings are top contributors, giving such insights as which feature to select for the project and how to measure the performance of such a model in the project.

In this project Vijn et al. (2020) will describe how feature selection and model evaluation methods have been used to build, and evaluate, predictive models in order to provide an analysis of stock price prediction via machine learning techniques.

List of the References

Kaur, J. and Dharni, K. (2022). Application and performance of data mining techniques in stock market: A review. *Intelligent Systems in Accounting, Finance and Management*, [online] 29(4), pp.1–17. doi:<https://doi.org/10.1002/isaf.1518>.

Peng, Y., Albuquerque, P.H.M., Kimura, H. and Saavedra, C.A.P.B. (2021). Feature selection and deep neural networks for stock price direction forecasting using technical analysis indicators. *Machine Learning with Applications*, [online] 5, p.100060. doi:<https://doi.org/10.1016/j.mlwa.2021.100060>.

Rouf, N., Malik, M.B., Arif, T., Sharma, S., Singh, S., Aich, S. and Kim, H.-C. (2021). Stock Market Prediction Using Machine Learning Techniques: A Decade Survey on Methodologies, Recent Developments, and Future Directions. *Electronics*, [online] 10(21), p.2717. doi:<https://doi.org/10.3390/electronics10212717>.

Singh, J. and Khushi, M. (2021). Feature Learning for Stock Price Prediction Shows a Significant Role of Analyst Rating. *Applied System Innovation*, [online] 4(1), p.17. doi:<https://doi.org/10.3390/asi4010017>.

Vijn, M., Chandola, D., Tikkiwal, V.A. and Kumar, A. (2020). Stock Closing Price Prediction using Machine Learning Techniques. *Procedia Computer Science*, [online] 167, pp.599–606. doi:<https://doi.org/10.1016/j.procs.2020.03.326>.

Appendix A –

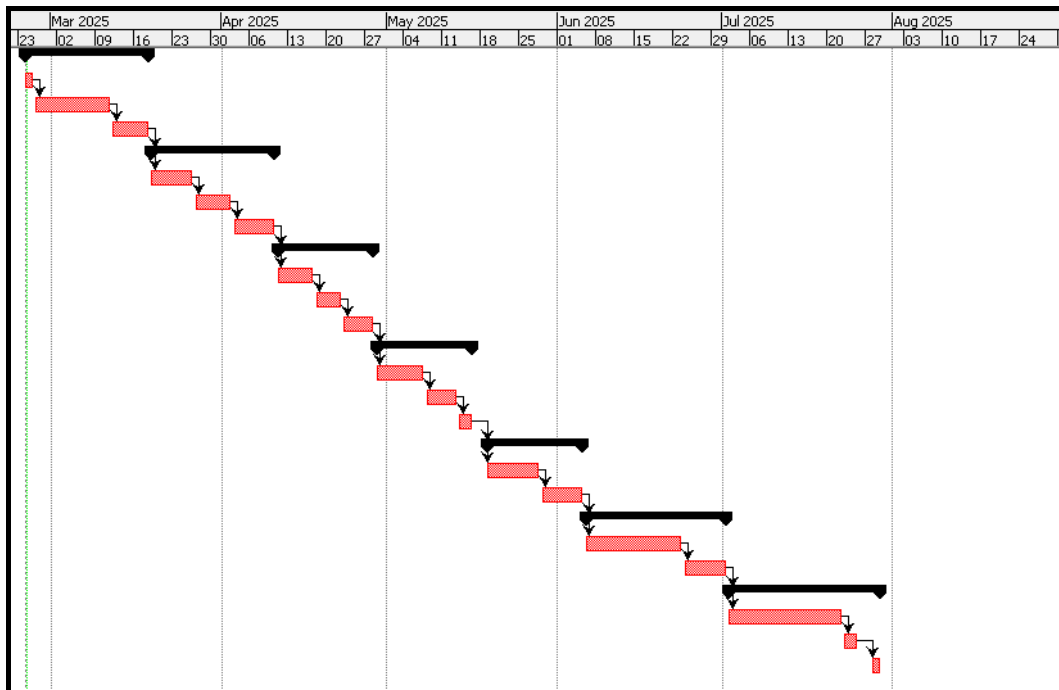


Figure 2: Gantt chart